AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 16 · STUDENT LESSON

Boundaries and 'Not Sure'

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Name the uncertainty. Request the right evidence.

Compare possible inputs, identify stable decisions and design review paths with explicit unresolved endings.

Keep one habit: Keep possibilities visible → compare decisions → seek evidence that resolves the actual gap.

MAIN LESSON · SECTION 1

What is uncertain, and would it change the decision?

> Name the evidence gap and request a check that addresses it.

Start with two possible inputs

An exact measurement can be unclear while the model's category stays unchanged. Compare the predictions under every stated possibility before deciding what needs review.

Retrieve the smudged-boundary question

A model predicts tall at a height of at least 6 cm. What would it predict if an unclear record could mean either 5 or 6 cm? Does a good result on a small final test settle this particular **input**?

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

By the end, you can...

- I can identify a specific source of **uncertainty** and explain how it affects a **prediction**.
- I can compare **prediction**s across stated **possible input**s without inventing a missing value.
- I can design a review step with relevant **evidence** and an explicit **unresolved** outcome.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

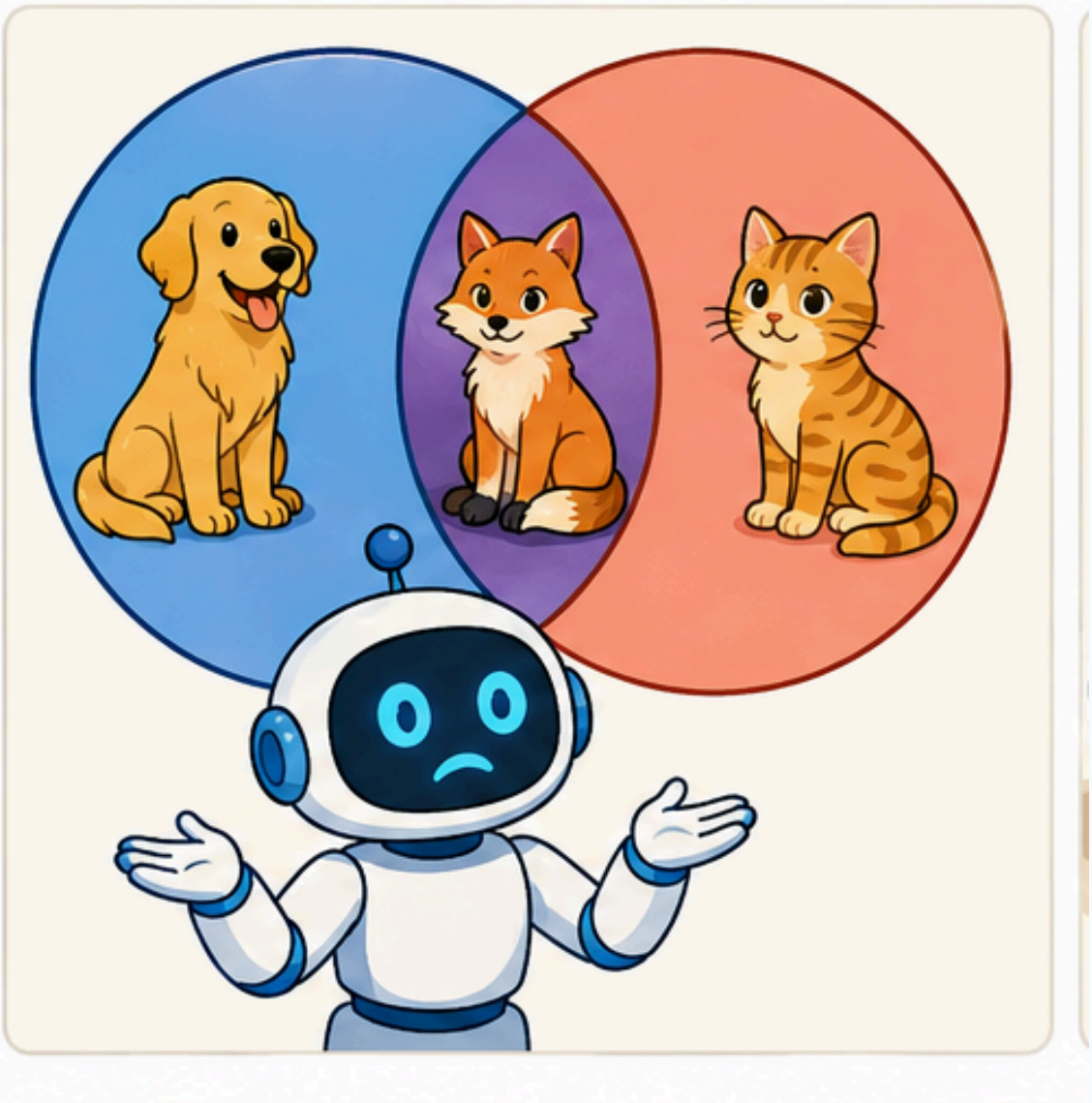
Remember before reading

Without opening the old lesson, explain one key idea from Save New Examples for Testing. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Learning from Examples. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from If/Then Decisions. Give an example and a mistake that the idea helps you avoid.

Overlapping categories create **uncertainty**; **abstain**ing can be responsible.



LEARNING TARGET A sorter receives a clear purple card, a red card hidden by a sleeve and a card labelled both red and blue by two reviewers. Give a different reason for **uncertainty** in each case and a suitable next action.

Learner name:

Readiness for the connected explanation

This week's end goal

By the end, I can use **uncertainty** to request a specific review when **possible input**s lead to different decisions.

MAIN LESSON · SECTION 2

Give uncertainty a specific reason

> A repeated calculation cannot create a missing measurement.

Distinguish different evidence gaps

An unclear measurement, a missing feature and a weakly tested model create different problems. Saying the AI is unsure does not identify which one matters.

Write what is known, what remains unknown and how the unknown could change the result. Then choose a check that addresses that specific gap.

Read the tower model and unclear input

Use the familiar tower model: tall at a height of at least 6 cm, short otherwise, for whole-number heights from 1 through 10. A smudged card could read 5 cm or 6 cm. The threshold is known; the **input** is unclear.

Repeating the same calculation cannot recover the missing digit.

State the complete-set assumption

For this paper investigation, treat the listed possibilities as the complete set allowed by the exercise. In real use, that assumption would itself need **evidence**.

We will compare the model's **prediction**s under each possibility; we are not changing Tower1 to accept an invented exact measurement.

Choose a relevant check

- Is the uncertainty about the measurement, the rule or evidence of model correctness? Which information is known in the 5-or-6 cm case, and what could resolve its missing digit?

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY A soft toy shaped like a ball may fit toy, sports, or comfort-item bins depending on the sorter's purpose.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

Compare predictions under the stated possibilities

> Keep uncertainty about the input separate from whether the output changes.

Read the three evidence situations

Possible heights can change the prediction

AVAILABLE RECORD	PREDICTIONS TO COMPARE	WHAT CAN BE STATED
Either 5 or 6 cm	5 gives short; 6 gives tall	Category unresolved
Either 8 or 9 cm	Both give tall	Tall under both inputs
No height supplied	Possibilities not listed	Request a measurement

A shared prediction across the listed possibilities does not identify the exact input or prove the category correct. Different predictions show that the unknown input can change this model's result.

Trace 5 or 6 across the boundary

At either 5 or 6 cm, one possibility predicts short and the other tall. Record no single category yet and ask for the original record or a new measurement. Do not select the more convenient digit.

Averaging the two readings would introduce a different **input** that neither possibility stated.

Limit the stable prediction for 8 or 9

At either 8 or 9 cm, the model predicts tall in both cases. You can state that the **prediction** stays tall under these two possibilities. Keep the exact-height **uncertainty** visible.

This agreement does not check the model against a correct task label or establish that the list of possible heights is complete.

Explain one comparison

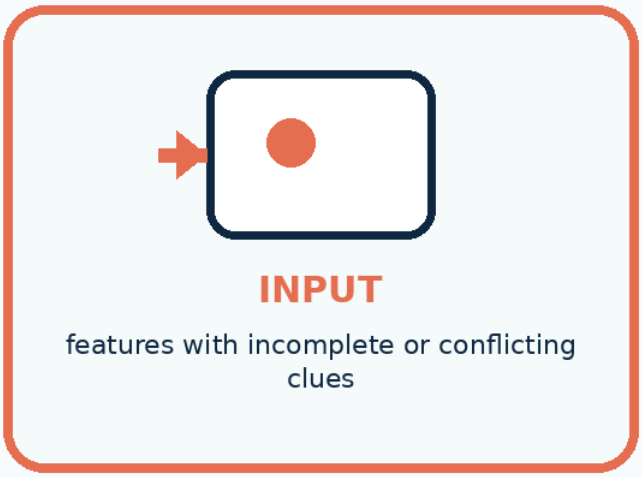
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

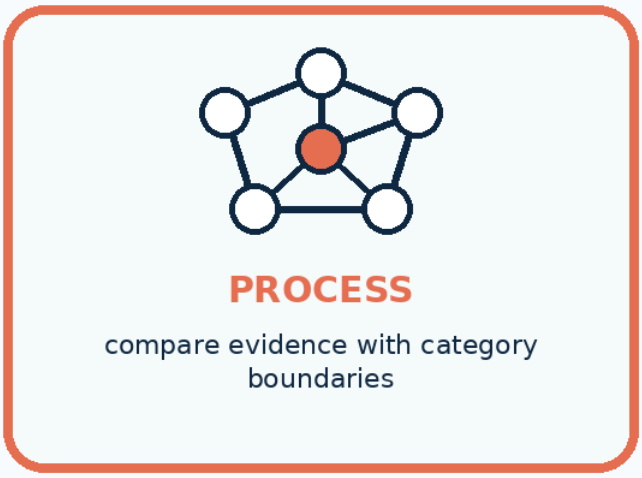
VISUAL **MECHANISM**

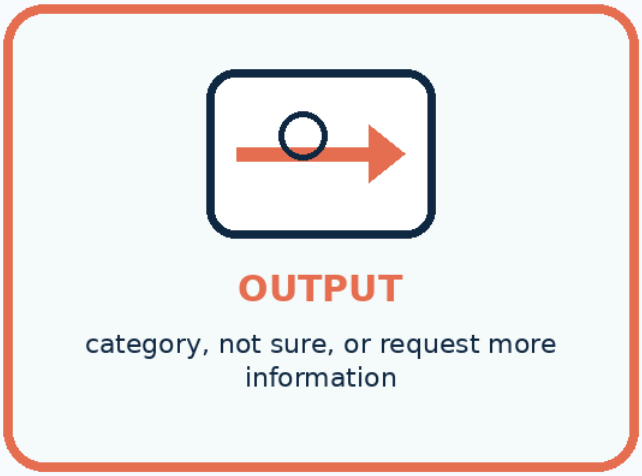
Input → Process → Output

Follow the information instead of imagining magic inside the machine.



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INPUT features with incomplete or conflicting clues

PROCESS compare **evidence** with category boundaries

OUTPUT category, not sure, or request **more information**

CORE IDEA Overlapping categories create **uncertainty**; **abstain**ing can be responsible.

UNDER THE HOOD Abstention is a decision policy: when confidence or **evidence** fails a threshold, the system defers.

Good thresholds balance coverage against error cost and specify what additional information or human review follows.

Draw a fresh example with all three stages:

A closer explanation

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare different reasons for review

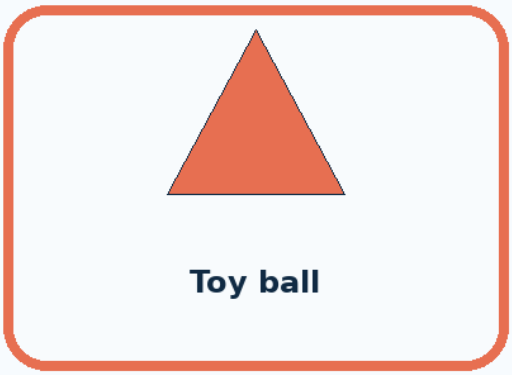
> Distinguish a missing observation, a category outside the guide and conflicting recorded labels.

Pictures paired with their evidence



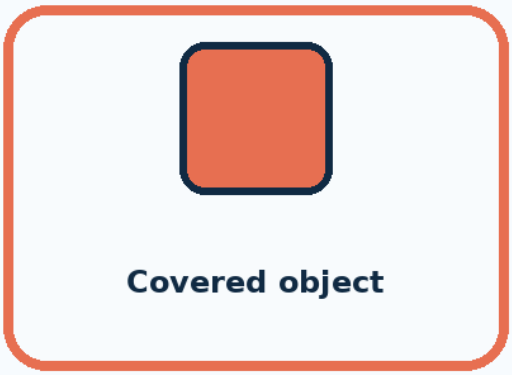
The case to investigate

CASE 1 — Clearly a book Clue: strong single-category **evidence**. Meaning: classify.



The case to investigate

CASE 2 — Toy ball Clue: toy and sports overlap. Meaning: review purpose.



The case to investigate

CASE 3 — Covered object Clue: important features hidden. Meaning: need more info.



The case to investigate

CASE 4 — Personal item Clue: high consequence if wrong. Meaning: adult review.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

BOUNDARY Responsible **uncertainty** is informative and actionable, not a blank refusal.

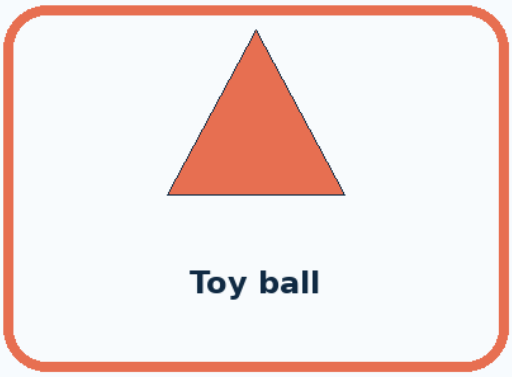
▼ Earlier wording, alternatives and picture transcripts

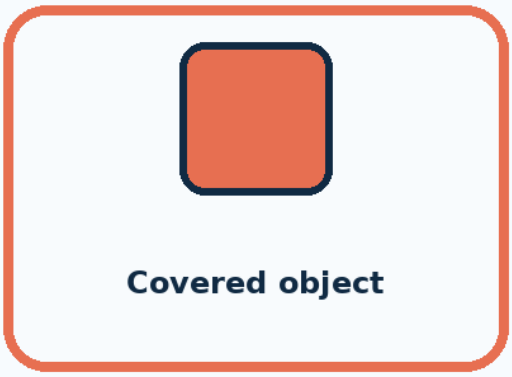
PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Clearly a book Clue: strong single-category **evidence**. Meaning: classify.

CASE 2 — Toy ball Clue: toy and sports overlap. Meaning: review purpose.

CASE 3 — Covered object Clue: important features hidden. Meaning: need more info.

CASE 4 — Personal item Clue: high consequence if wrong. Meaning: adult review.

Change one thing in the pictured case

Change: Change B's possibilities to 6 or 7 cm.

Prediction: Both fail the width limit; rejection no longer depends on resolving their difference.

Reason: All remaining possibilities lead to the same decision.

MAIN LESSON · SECTION 6

An unclear number need not imply an unclear decision

> The cubby rule accepts width at most 5 cm, and these exercises supply complete possibility sets.

The cubby rule and two width sets

A fictional cubby selector accepts a block if width is at most 5 cm.

An unclear measurement means Block A is either 4 or 5 cm; Block B is either 5 or 6 cm.

These complete possibility sets contain all widths under consideration. Other dimensions are known to fit.

Compare the possible decisions

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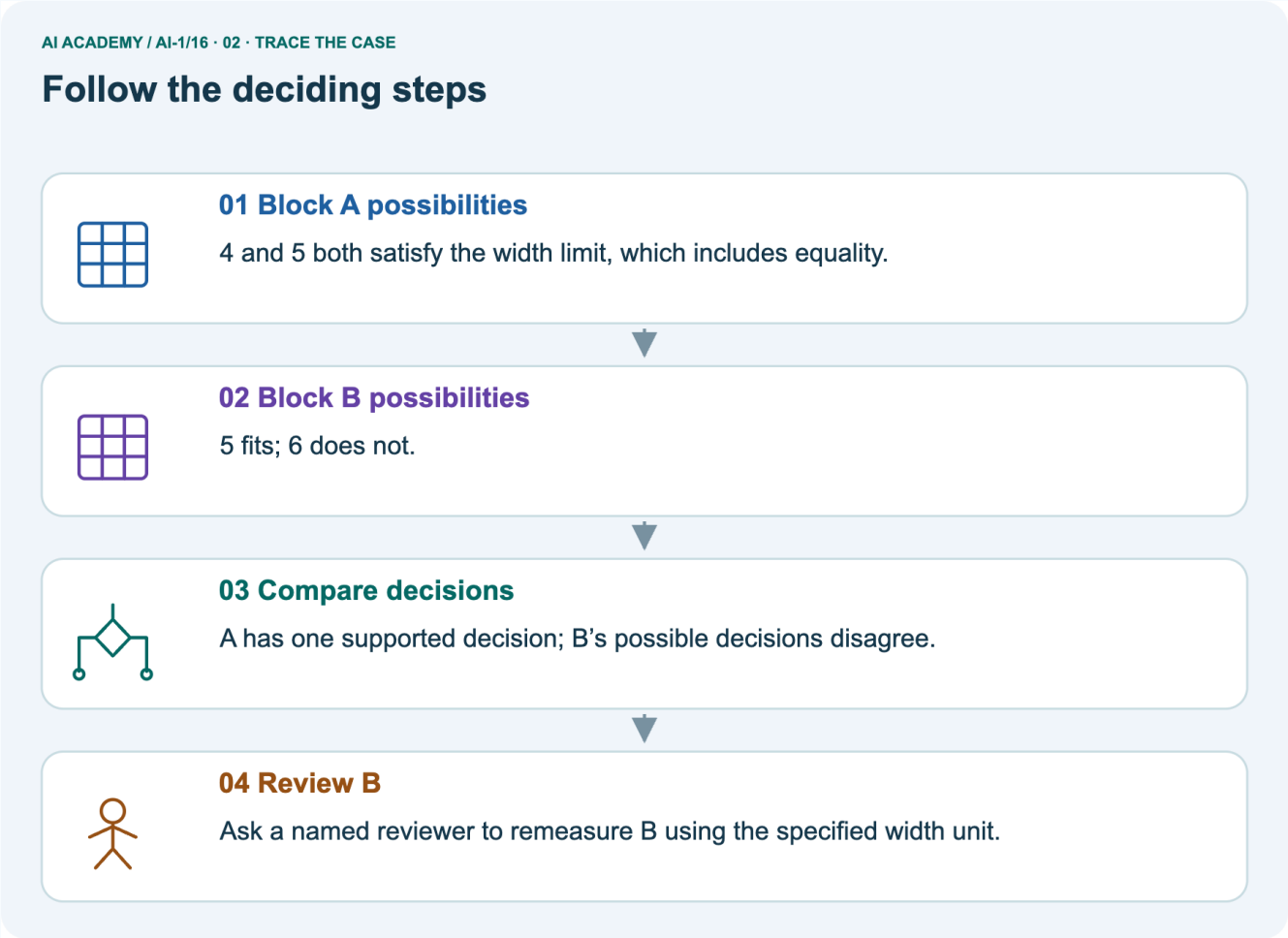
Possibilities and decisions

	Width 1 cm	Decision 1	Width 2 cm	Decision 2
Block A	4	fits	5	fits
Block B	5	fits	6	fails

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

Trace why B needs a measurement check



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain decision stability and disagreement

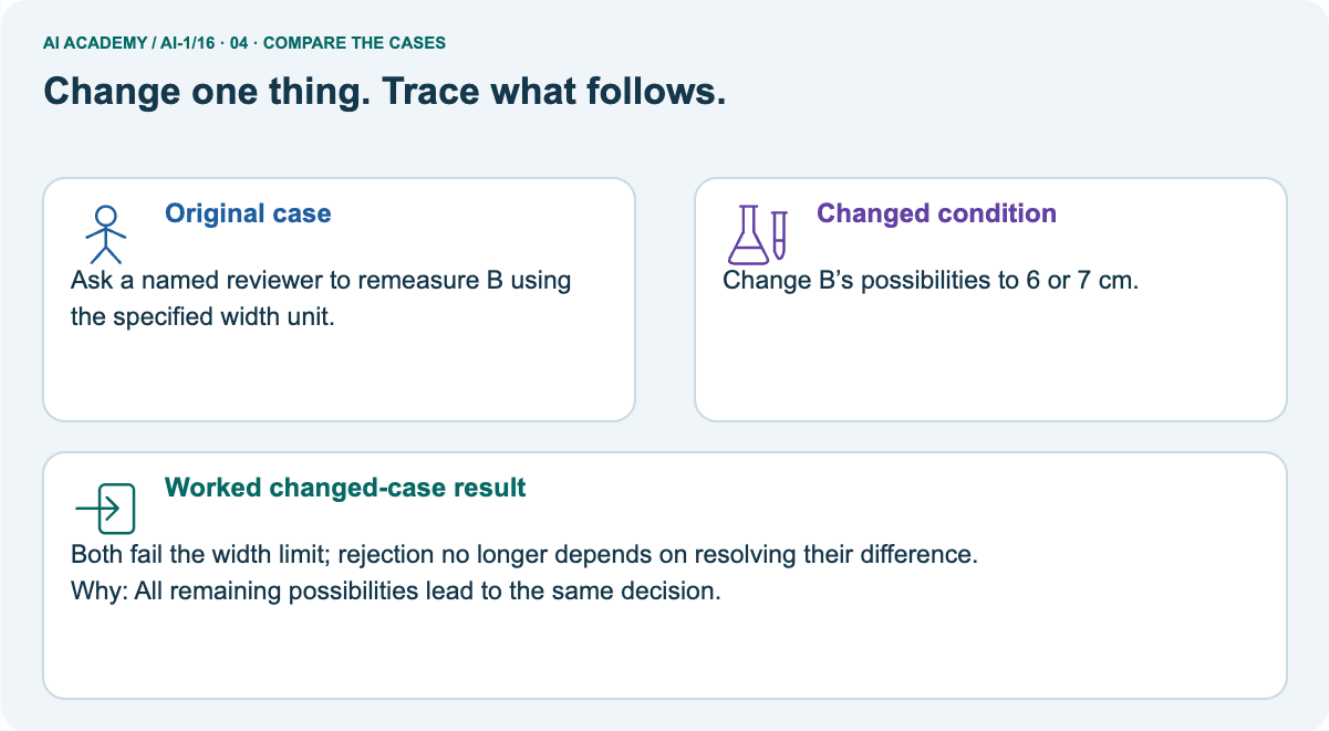
Uncertainty about a number need not always create **uncertainty** about the decision. Every allowed width for A fits, so its width **uncertainty** does not affect acceptance. B straddles the boundary.

Record both possibilities and request a better measurement instead of selecting the one that makes a preferred outcome.

Predict with a changed complete set

- For a changed hypothetical case, replace B's complete listed width possibilities with 6 or 7 cm. Predict the acceptance decision under each. Keep the at-most-5 limit and the other dimensions unchanged.

▼ Check the changed case



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the reviewer's evidence relevant

A review path is not automatically accurate; its conclusion depends on relevant, better **evidence**.

Specify the check B needs

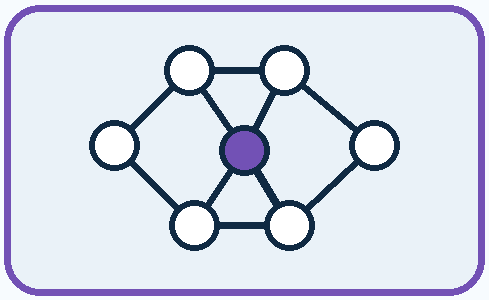
- Read the labelled picture: Which original block needs a better width before accepting or rejecting it? What should the reviewer do?

▼ Earlier wording, alternatives and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE A soft toy shaped like a ball may fit toy, sports, or comfort-item bins depending on the sorter's purpose.

STEP 1 — OBSERVE features with incomplete or conflicting clues

STEP 2 — TRACE compare **evidence** with category boundaries

STEP 3 — CONCLUDE category, not sure, or request **more information**

MODEL REASONING A useful not-sure result is specific: it states which categories compete, what **evidence** is missing, and who or what should supply it.

Abstention reduces forced errors but should not become an excuse to avoid all decisions.

DO NOT OVERCLAIM Responsible **uncertainty** is informative and actionable, not a blank refusal.

What new **evidence** could change this answer?

A closer explanation

Reason for review	Relevant check	What it cannot settle alone
Smudged height	Read the original or remeasure	Whether the model is accurate
Unit missing	Check how the record was made	The unit from digits alone
Two labels for one record	Check the label policy and sources	Truth by majority vote
Poorly tested new use	Evaluate relevant reserved cases	Every future outcome

A review request should carry the original **input**, model version, **unresolved** issue and any **prediction**s already made. Name the **evidence** to seek and the person responsible for checking it.

A reviewer needs access to relevant **evidence** and a clear task; being human does not make an unsupported guess correct.

After review, record what was checked and what changed. If a reliable new reading establishes 6 cm, the model predicts tall. If the record is still unreadable, keep no **prediction** with reason **unresolved** height.

Do not replace the original record without retaining the reason for the correction.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Boundaries and 'Not Sure'

The facts we will use

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Possibilities and decisions

	Width 1 cm	Decision 1	Width 2 cm	Decision 2
Block A	4	fits	5	fits
Block B	5	fits	6	fails

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

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Boundaries and 'Not Sure'

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Block A possibilities

4 and 5 both satisfy the width limit, which includes equality.



2. Block B possibilities

5 fits; 6 does not.



3. Compare decisions

A has one supported decision; B's possible decisions disagree.



4. Review B

Ask a named reviewer to remeasure B using the specified width unit.

CHANGE ONE THING

Narrow B's possibilities to 6 or 7 cm.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Block A possibilities	4 and 5 both satisfy the width limit, which includes equality.
2. Block B possibilities	5 fits; 6 does not.
3. Compare decisions	A has one supported decision; B's possible decisions disagree.
4. Review B	Ask a named reviewer to remeasure B using the specified width unit.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing: Change B's possibilities to 6 or 7 cm. Predict the result before checking the explanation. What remains the same?

Repeated agreement does not reveal a missing input

> State a supported prediction or unresolved outcome without inventing a confidence percentage.

Inspect what identical model runs actually repeat

A deterministic model repeats the same **prediction** for the same **input** and settings. Ten identical runs do not give ten independent measurements of a smudged digit.

Several models can also share a feature gap, **training** examples or mistaken assumption. Agreement alone is not proof that the **prediction** is correct.

Use a precise, limited result statement

Keep the claim specific: tall under both listed heights, **input** still unclear, or no category until the measurement is checked. Do not attach a made-up confidence percentage.

Evidence about correctness and the meaning of confidence will be developed in Weeks 17 and 18.

Repair the repeated-run claim

- CLAIM A A good system must always choose one label.
- CLAIM B Not sure means the system failed completely.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A learner runs the same model ten times on a guessed height and gets tall every time. They say review is no longer needed. Explain what the repetitions establish and name the evidence still needed

▼ **Earlier wording, alternatives and picture transcripts**

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Clearly a book	strong single-category evidence	classify
Toy ball	toy and sports overlap	review purpose
Covered object	important features hidden	need more info
Personal item	high consequence if wrong	adult review

Repair two tempting claims

Repair:

Repair:

A closer explanation

Practise: write a review request with two possible endings

> A useful review record preserves the original evidence and can end unresolved.

Use the explicit review policy

Use a written review policy: if stated possible valid heights lead to different **prediction**s, pause the single-category **output** and request a measurement check.

If a required **input** or its unit is missing, request that information. After the check, apply the model only to a supported valid **input**; otherwise retain an **unresolved** outcome.

Write the request and both endings

> **A card could mean 5 or 6 cm. Write the record sent for review, the check requested and two possible endings: a new measurement establishes 6 cm, or no reliable height can be obtained. Explain what each ending permits you to report**

Record unresolved evidence rather than repeating a failed request forever

A review path needs an ending even when **evidence** cannot be obtained. Repeating the same unsuccessful request forever does not resolve the **uncertainty**.

Record that the case remains **unresolved** and follow the agreed next step, such as holding the classroom sorting decision until the missing record is supplied.

Use W2 to limit a stable prediction

Use workbook W2 for the connected example “**Uncertainty** and review”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge “two possibilities means equal probabilities”

- Use the worked case for **Uncertainty** and review. Proposed explanation: “Two possible values imply equal probabilities”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Specify the pictured reviewer’s task

Which original block needs a better width before accepting or rejecting it? What should the reviewer do?

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Overlapping categories create **uncertainty**; **abstain**ing can be responsible.

Use a fresh example to explain **boundary case**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: Not sure means the system failed completely.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Practise with less help: Keep a stable prediction within its limits

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate actionable review paths

> Use three complete boundary sets and one missing-unit record to distinguish the evidence needed.

The rule, assumptions and case cards

Use CrateC, a fictional paper rule accepting width at most 4 cm, with other dimensions known to fit. Each listed possibility set is complete for this exercise: A={2,3} cm, B={4,5} cm, C={6,7} cm. For D, the recorded digits are 3 but the unit is missing. Compare every stated possible width before issuing one decision; request review when decisions differ or required input meaning is missing.

Compare, request a check and trace both endings

1. Predict the outcomes for A,B,C,D and name the exact reason for each review. For B, assign a measurement-checker role and request a supported width in cm. Trace both supplied fictional endings: a checked record establishes 5 cm, or the record remains unreadable. Keep original possibilities, check, resulting decision or unresolved reason together.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Keep uncertainty and outcome visible

Case	Prediction	Observed result	What it means
Normal			
Edge			

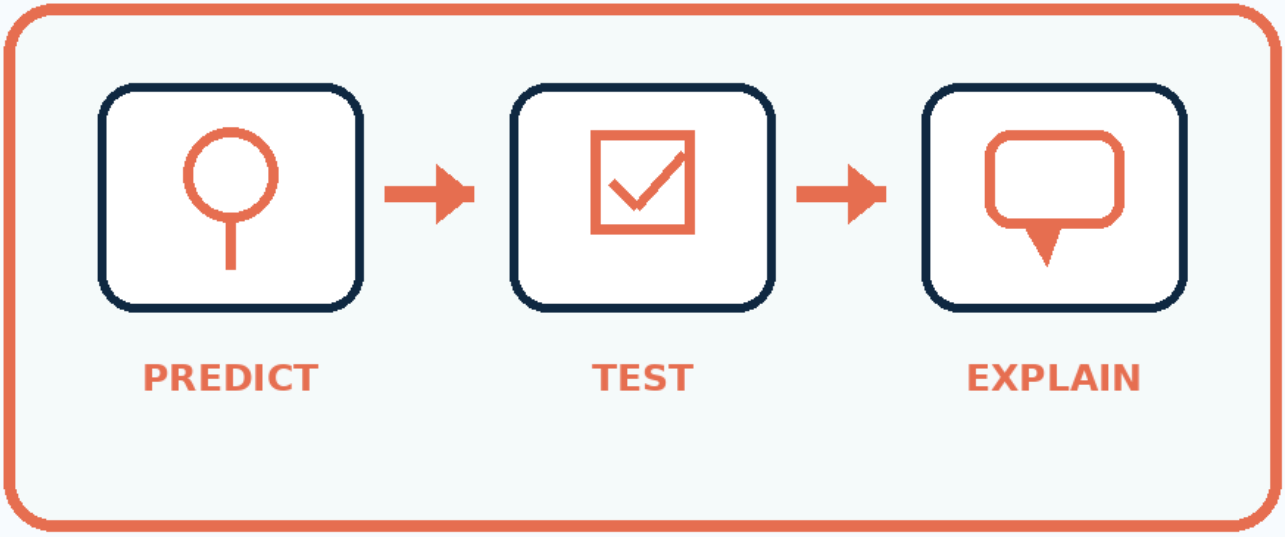
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EXPLORER LAB

Boundary and Abstention Studio

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Find three **boundary case**s and design a 'need more info' **output**.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: transfer uncertainty handling to distance

> Compare every input in each complete set under the supplied near/far model.

The new model and possibility sets

A paper model predicts far at a distance of at least 4 units, otherwise near, for whole-number distances 1 through 8. For each complete possibility set, compare **prediction**s: 2 or 3; 3 or 4; 4 or 5.

Identify the set requiring review before a single category can be stated. Design a relevant check and an **unresolved** ending. Limit the claims for the other two sets.

> **My independent explanation, trace and limits**

Keep one first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Transfer the method to distance

Week 16 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A sorter knows pencils and books. You show a lunch box. Should it force a known label or ask for help? State a useful fallback.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Add a not-sure branch, missing-information prompt, and human review destination to the sorter.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A sorter receives a clear purple card, a red card hidden by a sleeve and a card labelled both red and blue by two reviewers. Give a different reason for **uncertainty** in each case and a suitable next action.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Give each review branch a reason and an ending

> Name the evidence to seek and the role responsible for checking it.

This week’s project decision

THIS WEEK'S CHECKPOINT Add a not-sure branch, missing-information prompt, and human review destination to the sorter.

Add explicit review paths to the sorter, including **unresolved input**, responsible person and **evidence** needed.

Plan inputs, checks and unresolved outcomes

- What real classroom need does this serve?
- What information is necessary—and what should stay private?

- What happens inside the **process**?
- What normal and edge cases will you test?
- When should it say not sure or ask a person?

Make uncertainty informative and actionable

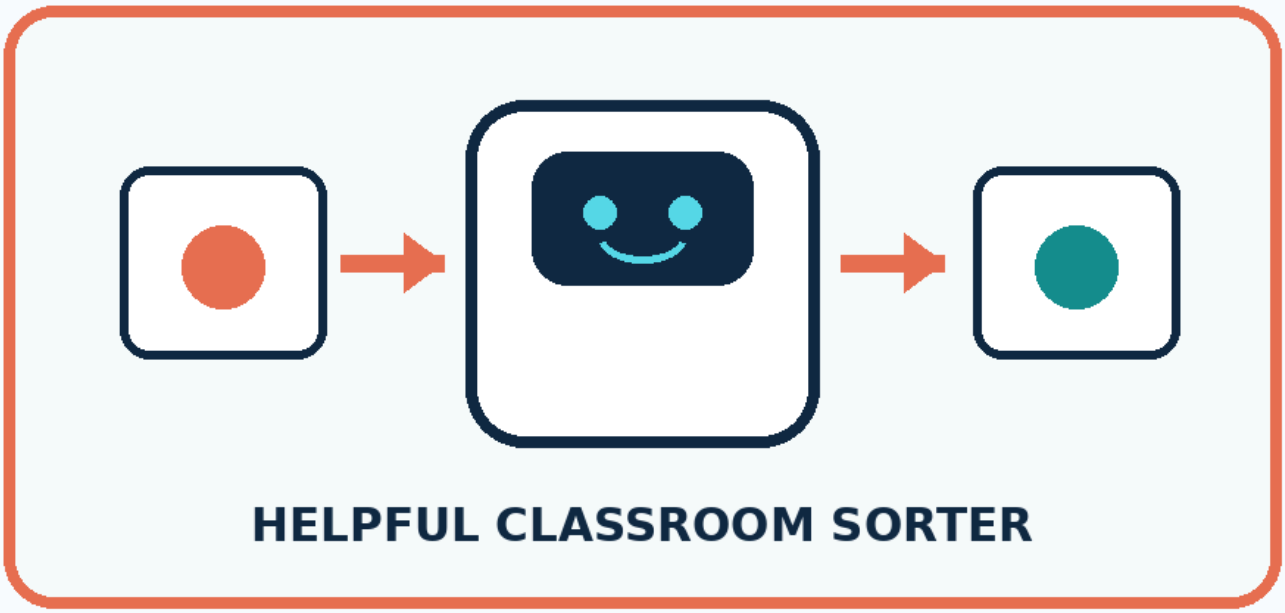
RESPONSIBLE DESIGN Responsible **uncertainty** is informative and actionable, not a blank refusal.

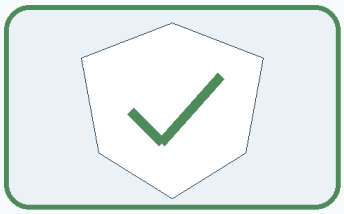
▼ **Earlier wording, alternatives and picture transcripts**

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: state what remains unknown

> Close your notes and distinguish a stable model prediction from checked input or model correctness.

Your exit explanation

- Close your notes. Explain this goal in your own words: Identify a specific source of **uncertainty** and explain how it affects a **prediction**

> **My deciding step, evidence and one remaining question**

Choose support and revisit later

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
boundary case	example near competing category rules	
uncertainty	limited support for one conclusion	
abstain	decline to choose a category	
escalation	send the case for added information or review	

Terms for the connected explanation

Word	Meaning
Possible input	An input still compatible with the stated information
Review trigger	A stated condition that sends a case for an additional check
Review evidence	Information gathered to investigate the reason a case was referred

Word	Meaning
Unresolved	Still not settled after the available checks

Repair a claim using a word

- CLAIM TO REPAIR A good system must always choose one label.
- Use one vocabulary word and one piece of **evidence** to repair it:

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ Choose a stronger thinking challenge

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

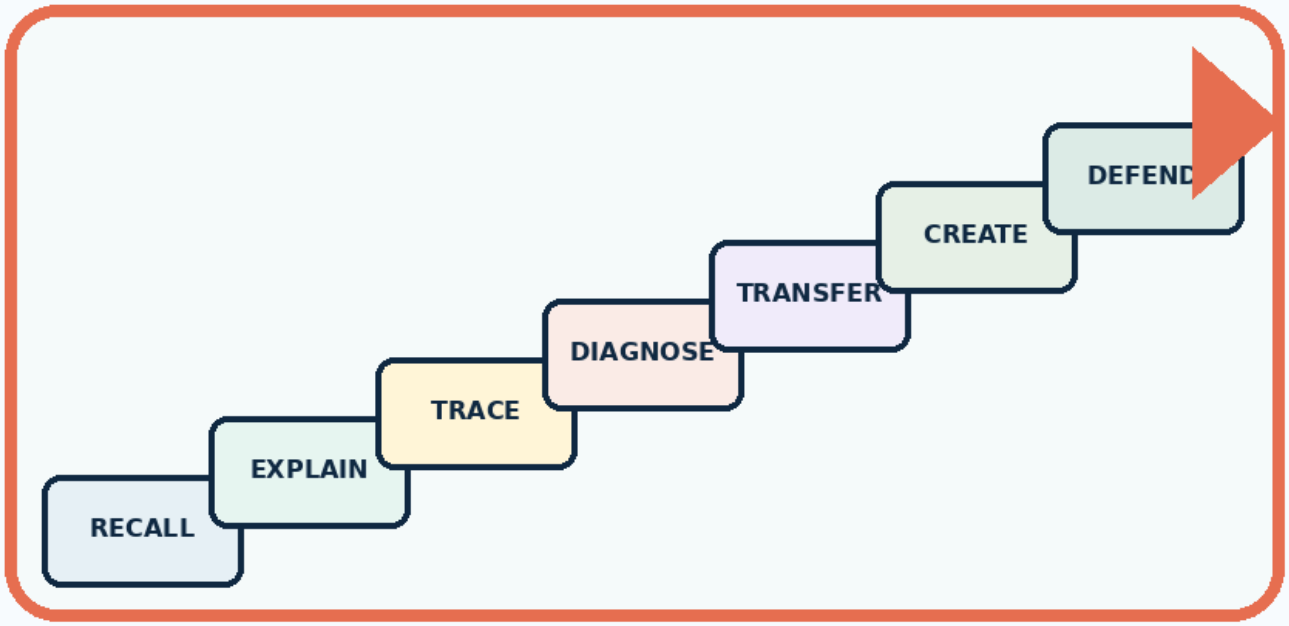
- RECALL Define **boundary case**.
- EXPLAIN Overlapping categories create **uncertainty**; **abstain**ing can be responsible.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: A good system must always choose one label.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Transfer the method to distance is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

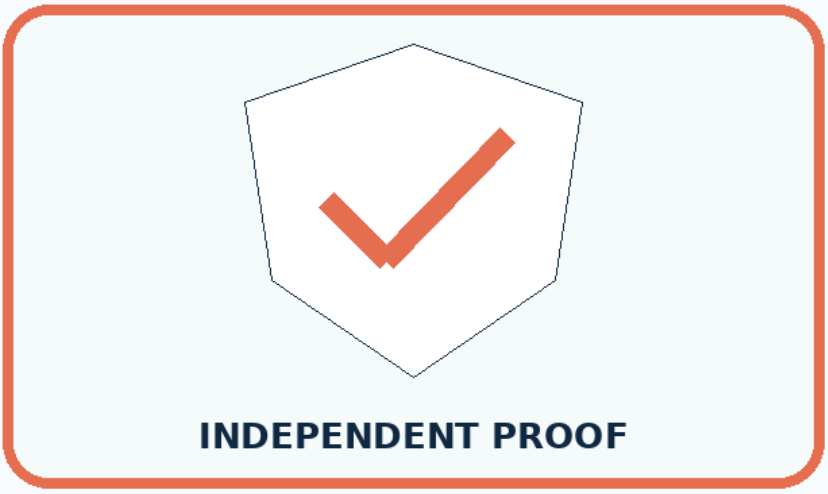
My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ Earlier wording, alternatives and picture transcripts

MASTERY

Checklist for explaining this week's idea

Use after practice. Transfer the method to distance is the selected independent case.



EXPLANATION CHECK Identify the specific evidence gap, compare all stated possibilities, request relevant review and retain an unresolved ending when evidence is still missing.

Independent **evidence**:

SELF-CHECK A useful not-sure result is specific: it states which categories compete, what **evidence** is missing, and who or what should supply it.

Abstention reduces forced errors but should not become an excuse to avoid all decisions.

NEXT BRIDGE Carry today's **evidence** habit into: **Prediction** Is Not Fact.

▼ **Extra practice: Two different reasons to ask for help**

REFERENCE · SECTION 13

Extra practice: Two different reasons to ask for help

> Try the question before opening its worked explanation.

Read the supplied case and question

A sorter handles clearly red and clearly blue cards. Purple and very dark cards arrive.

How should it handle each uncertain case?

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Use a review tray. Purple is outside the clear categories; darkness prevents **observation**.
These are different **uncertainty** causes.

Evidence or stage	Value or result
Purple card	Outside clear red/blue categories
Very dark card	Colour cannot be observed
Review tray	Preserve uncertainty for a person

▼ Earlier wording, alternatives and picture transcripts

Week 16 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Two different reasons to ask for help

A sorter handles clearly red and clearly blue cards. Purple and very dark cards arrive.

How should it handle each uncertain case?

Use a review tray. Purple is outside the clear categories; darkness prevents **observation**.
These are different **uncertainty** causes.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Identify a specific source of uncertainty and explain how it affects a prediction	Explain the deciding step, show a trace or test, and state one limit.
Compare predictions across stated possible inputs without inventing a missing value	Explain the deciding step, show a trace or test, and state one limit.
Design a review step with relevant evidence and an explicit unresolved outcome	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional cubby selector accepts a block if width is at most 5 cm.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional cubby selector accepts a block if width is at most 5 cm.
- Which labelled step matches this detail? Ask a named reviewer to remeasure B using the specified width unit.
- 1. Block B possibilities
- 2. Compare decisions
- 3. Block A possibilities
- 4. Review B
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Review B Ask a named reviewer to remeasure B using the specified width unit. **Uncertainty** about a number need not always create **uncertainty** about the decision.

Every allowed width for A fits, so its width **uncertainty** does not affect acceptance. B straddles the boundary.

Record both possibilities and request a better measurement instead of selecting the one that makes a preferred outcome.

Changed-case question

- Changed-case practice: Change B's possibilities to 6 or 7 cm. Which conclusion is supported by the supplied facts?
- 1. Both fail the width limit; rejection no longer depends on resolving their difference.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the changed case

After your attempt, compare with the supported explanation: Both fail the width limit; rejection no longer depends on resolving their difference. All remaining possibilities lead to the same decision.

Repair and save your work

Review Save New Examples for Testing (ai-1/15). Goal: Separate examples used to choose a model from examples reserved for its final evaluation. Short practice: Does 4/4 on the replay replace the original 3/4 result? Explain..

Return to this week and explain what you changed.

Review Learning from Examples (ai-1/13). Goal: Fit a threshold by tracing **prediction**s and counting disagreements for every candidate. Short practice: Which new measured length would reveal disagreement between the candidates?.

Return to this week and explain what you changed.

Review If/Then Decisions (ai-1/10). Goal: Evaluate a condition using the current state and trace only the selected branch. Short practice: Which **input** detects the difference between the two written conditions?

Explain its entire run.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

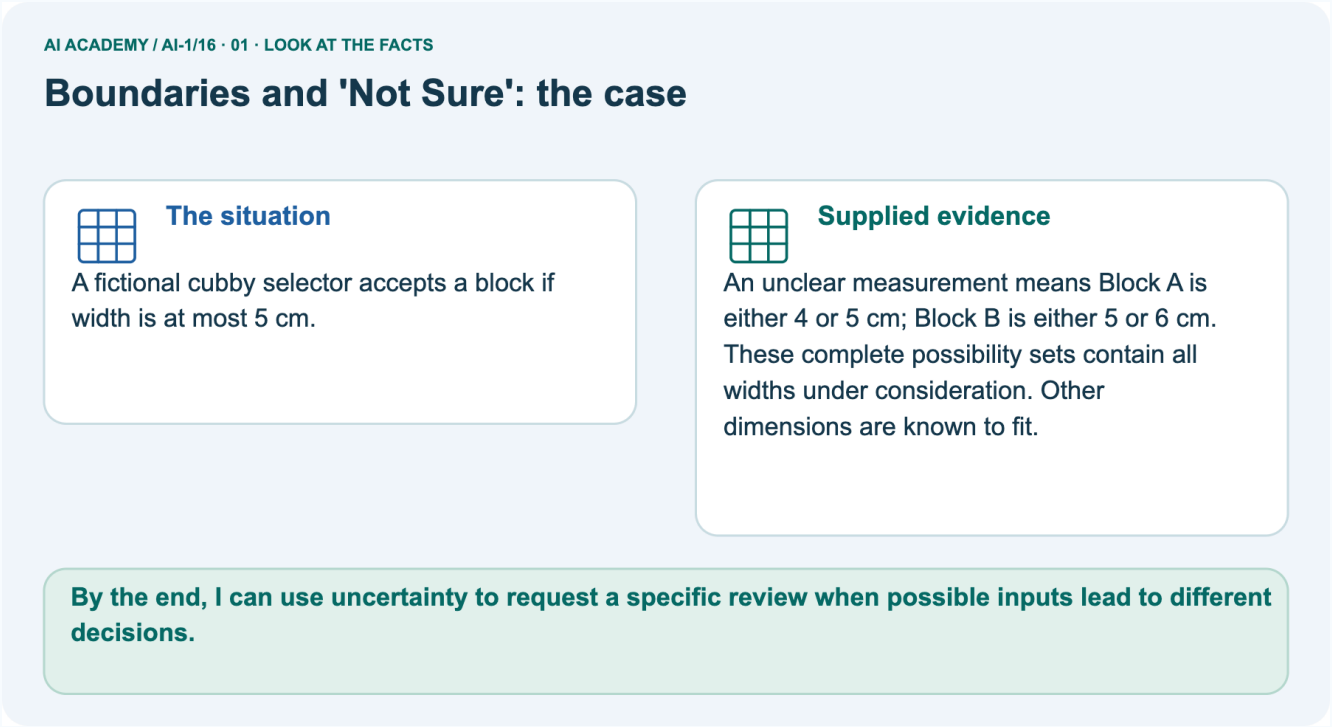
▼ Reference: the six-picture journey

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Boundaries and 'Not Sure': the case



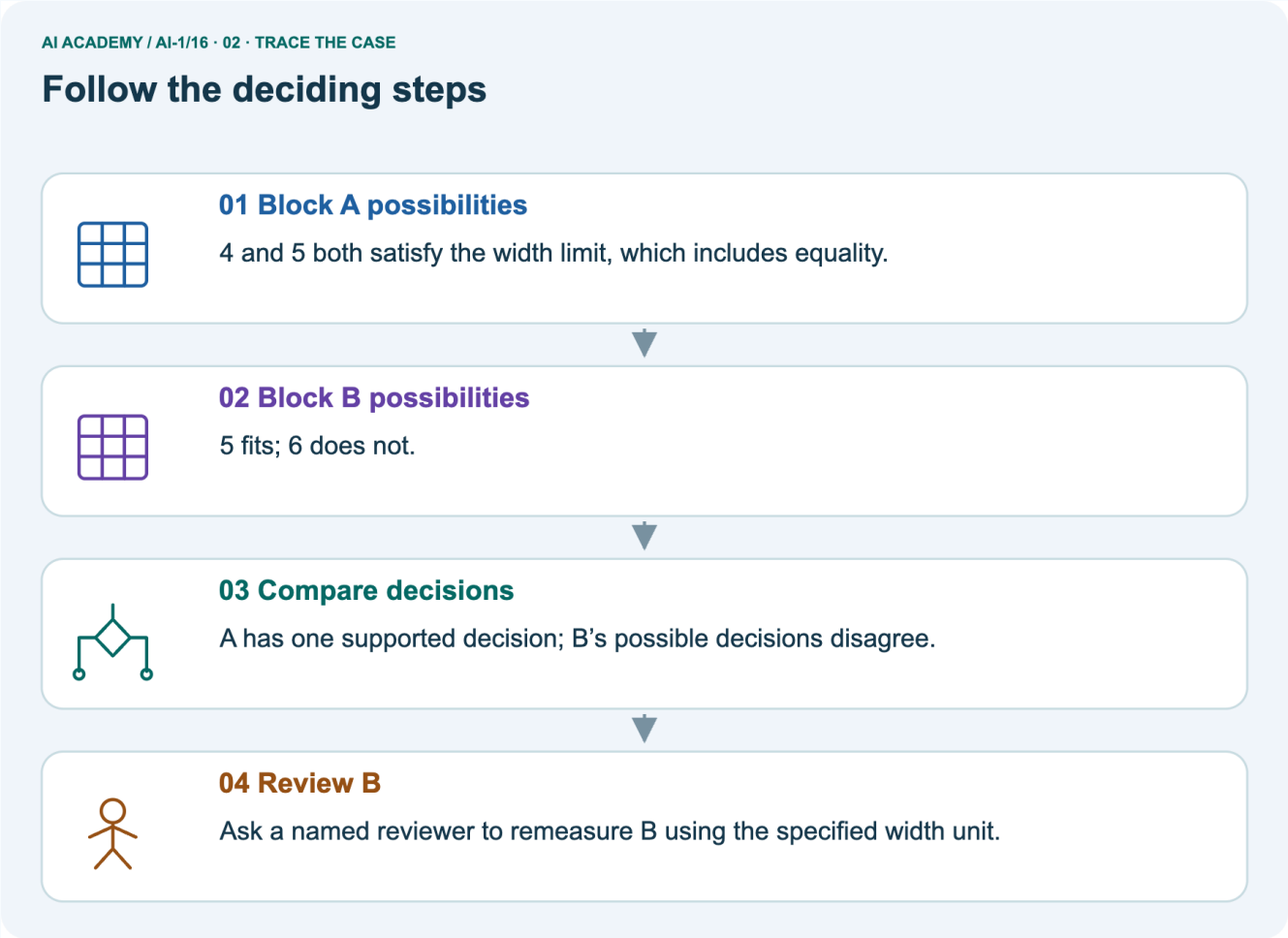
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional cubby selector accepts a block if width is at most 5 cm. Supplied facts: An unclear measurement means Block A is either 4 or 5 cm; Block B is either 5 or 6 cm.

These complete possibility sets contain all widths under consideration. Other dimensions are known to fit.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

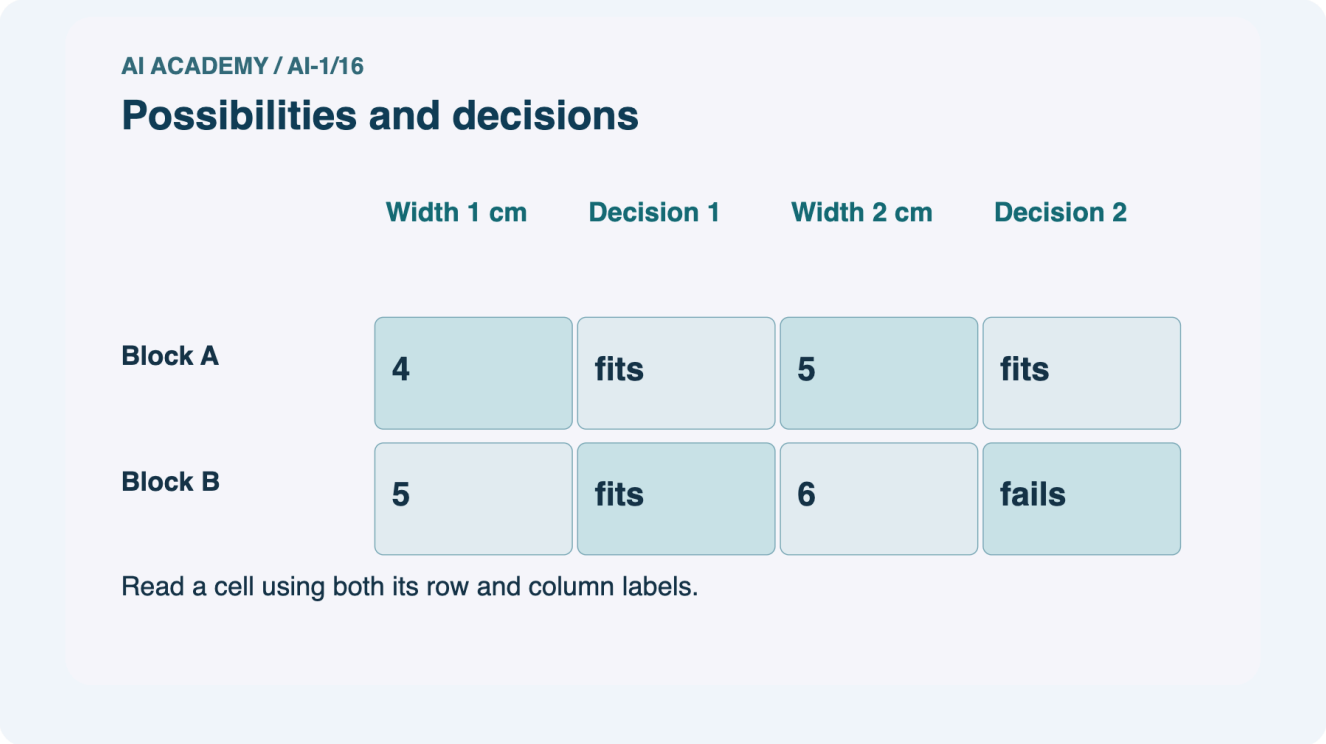


Follow the labelled picture with the matching explanation. Open the vector diagram

Block A possibilities: 4 and 5 both satisfy the width limit, which includes equality. → Block B possibilities: 5 fits; 6 does not. → Compare decisions: A has one supported decision; B's possible decisions disagree. → Review B: Ask a named reviewer to remeasure B using the specified width unit.

Point to each numbered stage and explain which recorded **input** it uses.

03. Possibilities and decisions

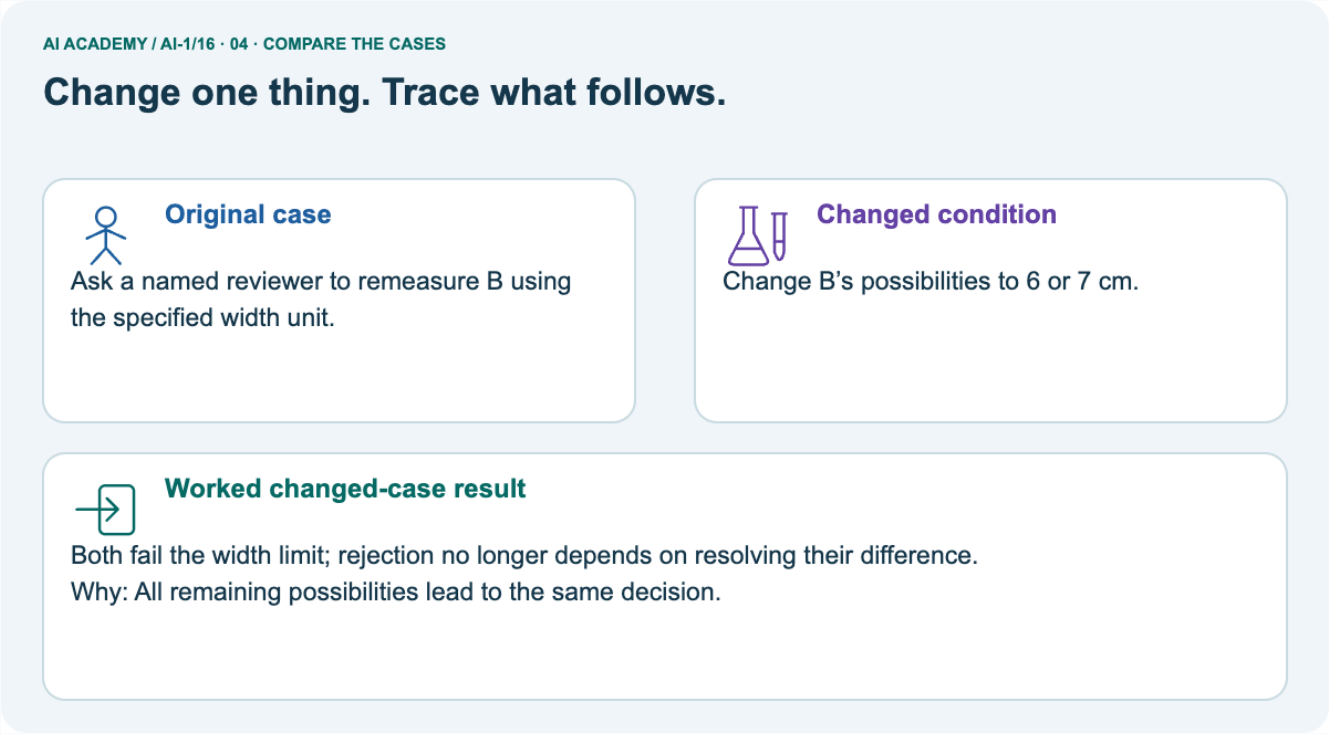


Follow the labelled picture with the matching explanation. Open the vector diagram

Possibilities and decisions. Block A: Width 1 cm = 4, Decision 1 = fits, Width 2 cm = 5, Decision 2 = fits. Block B: Width 1 cm = 5, Decision 1 = fits, Width 2 cm = 6, Decision 2 = fails.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



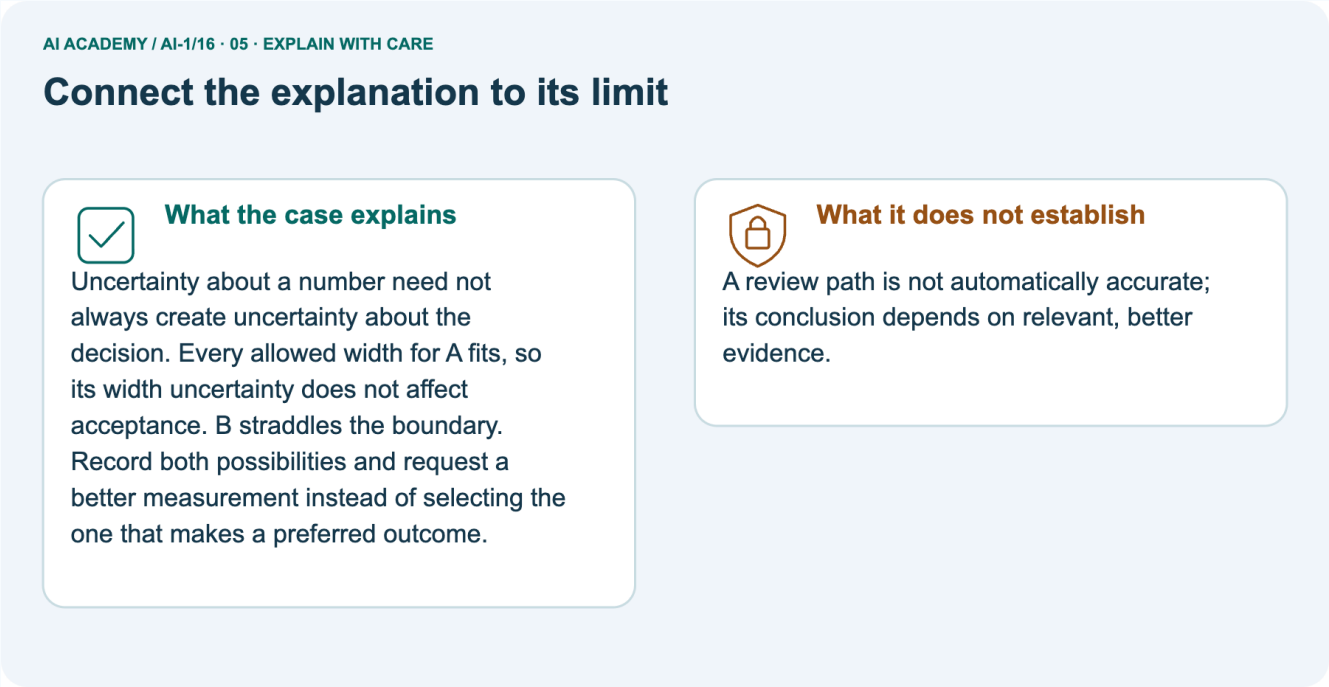
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Ask a named reviewer to remeasure B using the specified width unit. Changed condition: Change B's possibilities to 6 or 7 cm.

Expected result: Both fail the width limit; rejection no longer depends on resolving their difference. Reason: All remaining possibilities lead to the same decision.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



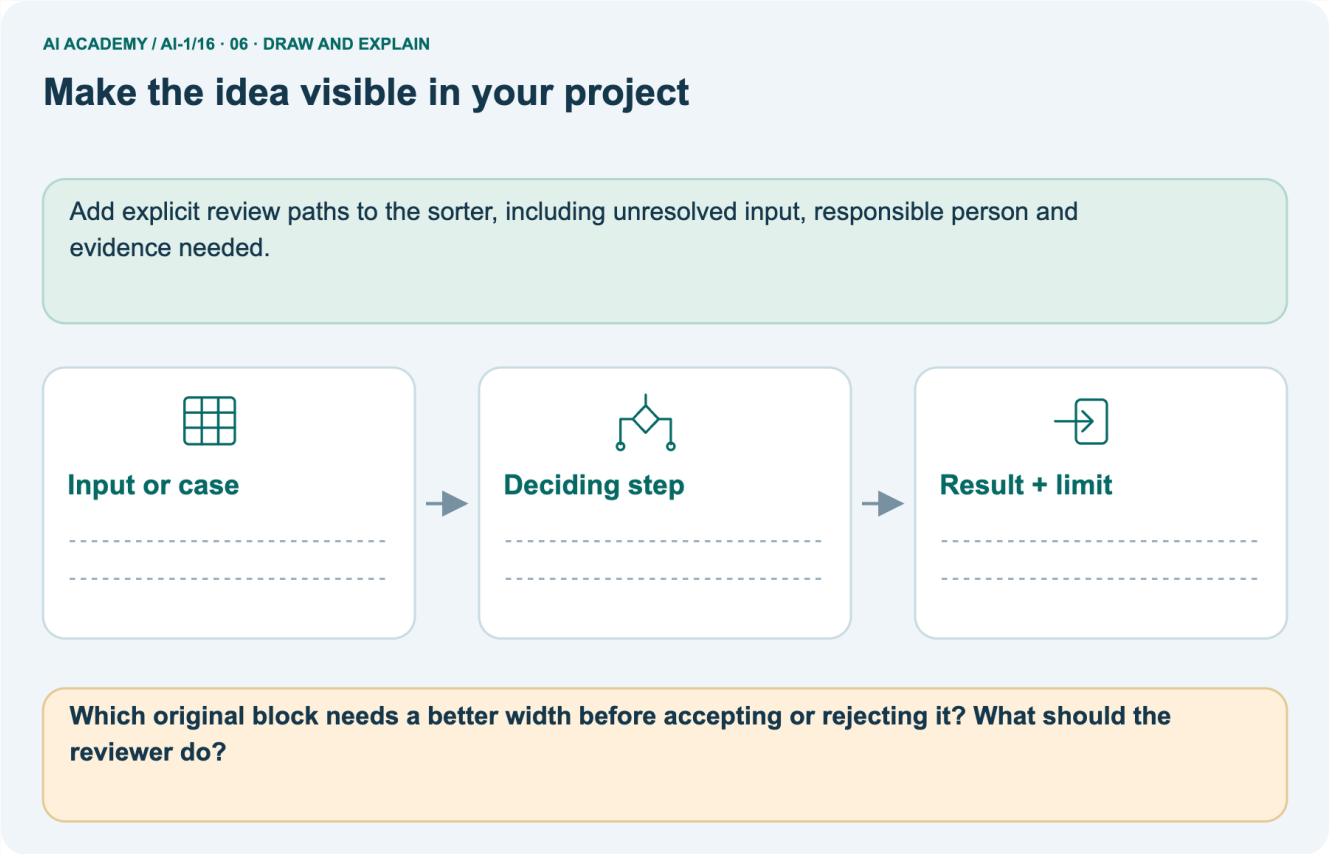
Follow the labelled picture with the matching explanation. Open the vector diagram

Uncertainty about a number need not always create **uncertainty** about the decision. Every allowed width for A fits, so its width **uncertainty** does not affect acceptance. B straddles the boundary.

Record both possibilities and request a better measurement instead of selecting the one that makes a preferred outcome. Limit: A review path is not automatically accurate; its conclusion depends on relevant, better **evidence**.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Add explicit review paths to the sorter, including **unresolved input**, responsible person and **evidence** needed. Draw your **input**, deciding step, result and limit.

Supported question: Which original block needs a better width before accepting or rejecting it? What should the reviewer do?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Boundaries and 'Not Sure': the case

02. Follow the deciding steps

03. Possibilities and decisions

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.